

Indian Institute of Engineering Science and Technology, Shibpur
Department of Computer Science and Technology

Algorithm Lab (CS2271)
Assignment 1

1. Write a C program that generates **n** random integers, where the value of **n** is provided by the user, and stores these integers in a file. Then, read all the elements from the file and display them on the screen. Finally, determine and print the **maximum** and **minimum** values from the list.
2. Write a C program to generate an array of **50,000** random elements. Implement both **Linear Search** and **Binary Search** algorithms to search for a given element in the array. If the element is found, display its position. Finally, compare the two search methods by reporting the **number of comparisons** performed and the **time taken** to locate the element.
3. Write a C program that systematically creates ten separate arrays with sizes of 5000, 10000, 15000, 20000, 25000, 30000, 35000, 40000, 45000, and 50000 elements, respectively. Populate each array with randomly generated integer values. For every array, implement and apply four different sorting algorithms i.e., Bubble Sort, Insertion Sort, Quick Sort, and Merge Sort to arrange the elements in ascending order. Measure and record the execution time taken by each sorting algorithm for each array size, and finally perform a detailed comparison of their time performance to analyze and evaluate the efficiency of these sorting techniques as the input size increases.